```
import pandas as pd
import re
import spacy
from collections import Counter
import matplotlib.pyplot as plt
```

```
df = pd.read_csv('/content/Tweets.csv')
df.head()
```

·line_sentiment	airline_sentiment_confidence	negative
neutral	1.0000	
positive	0.3486	
neutral	0.6837	
negative	1.0000	Ba
negative	1.0000	C

Next steps: Generate code with df New interactive sheet

```
def clean_tweet(text):
    text = re.sub(r"http\S+|www\S+", "", text)       # remove URLs
    text = re.sub(r"@\w+", "", text)                 # remove mentions
    text = re.sub(r"#", "", text)                    # remove hashtag symbol
    text = re.sub(r"[^\w\s]", "", text)              # remove emojis & punctuat
    text = text.lower().strip()
    return text

df["cleaned_text"] = df["text"].apply(clean_tweet)
df[["text", "cleaned_text"]].head()
```

	text	cleaned_text
0	@VirginAmerica What @dhepburn said.	what said
1	@VirginAmerica plus you've added commercials t...	plus youve added commercials to the experience...
2	@VirginAmerica I didn't today... Must mean I n...	i didnt today must mean i need to take another...
3	@VirginAmerica it's really aggressive to blast...	its really aggressive to blast obnoxious enter...
4	@VirginAmerica and it's a really big bad thing...	and its a really big bad thing about it

```
import spacy
nlp = spacy.load("en_core_web_sm")
```

```
from spacy.language import Language

@Language.component("hashtag_extractor")
def hashtag_extractor(doc):
    doc._.hashtags = [token.text for token in doc if token.text.startswith("#")
    return doc
```

```
from spacy.tokens import Doc

Doc.set_extension("hashtags", default=[])

nlp.add_pipe("hashtag_extractor", last=True)
```

hashtag_extractor
def hashtag_extractor(doc)

<no docstring>

```
docs =6 list(nlp.pipe(df['cleaned_text']))
```

```
data = []

for doc, sentiment in zip(docs, df['airline_sentiment']):
    for token in doc:
        if not token.is_stop and token.is_alpha:
            data.append({
                "token": token.text,
```

```
                "lemma": token.lemma_,
                "pos": token.pos_,
                "sentiment": sentiment
            })

ling_df = pd.DataFrame(data)
ling_df.head()
```

	token	lemma	pos	sentiment
0	said	say	VERB	neutral
1	plus	plus	CCONJ	positive
2	ve	ve	AUX	positive
3	added	add	VERB	positive
4	commercials	commercial	NOUN	positive

```
all_hashtags = []
for doc in docs:
    all_hashtags.extend(doc._.hashtags)

hashtag_df = pd.DataFrame(all_hashtags, columns=["hashtag"])
hashtag_df.value_counts().head()
```

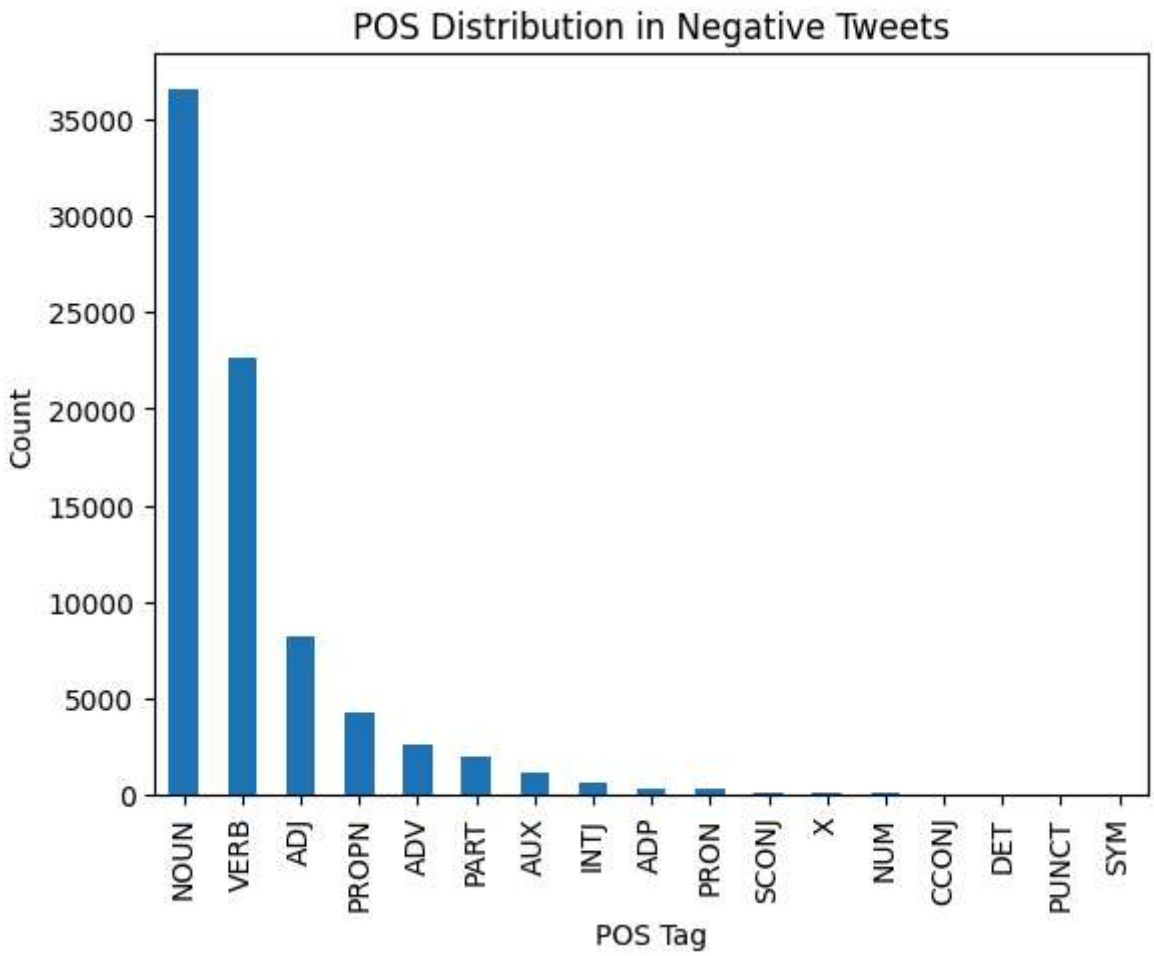
hashtag	count

dtype: int64

```
import matplotlib.pyplot as plt

top_hashtags = hashtag_df.value_counts().head(10)

# Check if top_hashtags is empty before plotting
if not top_hashtags.empty:
    top_hashtags.plot(kind='bar')
    plt.title("Top 10 Frequent Hashtags")
    plt.xlabel("Hashtag")
    plt.ylabel("Frequency")
    plt.show()
else:
    print("No hashtags found to plot.")
```

```
No hashtags found to plot.
```

```
neg_pos = ling_df[ling_df['sentiment'] == 'negative']['pos'].value_counts()

neg_pos.plot(kind='bar')
plt.title("POS Distribution in Negative Tweets")
plt.xlabel("POS Tag")
plt.ylabel("Count")
plt.show()
```